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Development of a catalyst Co-pyrolysis process for producing pyrolysis oil and wax from cooking oil contaminated polypropylene plastic

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The increase in food packaging waste, especially waste contaminated by cooking oil, has drawn increasing attention because of the difficulty in recycling this material. Waste-to-energy approaches are generally applied for using oily food packaging waste as solid fuel. Nevertheless, oily food packaging waste has the potential to be used as feedstock for higher-value products such as pyrolysis oil and wax. This study investigated the effects of catalysts and operating conditions of fast pyrolysis on the yield and properties of pyrolysis oil and wax produced from polypropylene plastic with palm oil as the representative oily food packaging waste. The presence of palm oil promoted wax formation but significantly increased the kinematic viscosity and acid value of pyrolysis oil. On the other hand, increasing the kaolin catalyst loading, pyrolysis temperature and heating rate reduced wax formation but did not improve the kinematic viscosity or acid value of the pyrolysis oil. The addition of activated carbon derived from rice husk as a cocatalyst effectively reduced the acid value of the pyrolysis oil and enhanced wax formation. The pyrolysis wax had a viscosity index greater than 400 and a pour point of 26 °C. The FTIR and GC-MS analyses revealed that the pyrolysis wax primarily consisted of saturated long-chain fatty acids and alcohols. Catalyst reusability testing demonstrated that there was no significant reduction in pyrolysis oil or wax yield across three consecutive cycles. Techno-economic evaluation indicated that co-producing pyrolysis oil and wax from cooking oil-contaminated PP plastic waste yielded more favorable outcomes than producing pyrolysis oil alone. A net profit of approximately 0.086 USD per kilogram of feedstock was achieved under the evaluated conditions.

Keywords Palm oil, Polypropylene, Pyrolysis oil, Pyrolysis wax, Co-catalyst

The surge in global plastic consumption over the past three decades has been remarkable, leading to the generation of approximately 129 million tons of plastic waste annually. Owing to its light weight, high durability, and cost-effectiveness, plastic waste has traditionally been managed through disposal methods such as landfilling and incineration. Unfortunately, improper management of these processes has resulted in significant environmental challenges, including air pollution and threats to human health[1].

While recycling efforts have been made for plastic waste, substantial challenges arise when dealing with plastics contaminated by food residues, particularly from the food delivery sector. The complexities involved in handling such waste, including the cost of drying material, controlling odors, and managing waste during processing, often render it unsuitable for recycling. As a consequence, this type of plastic waste is frequently directed to landfills, contributing to environmental degradation[2].

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In the context of plastic waste management in Thailand, polypropylene (PP) bags, a prevalent plastic type, are frequently contaminated with palm oil used in cooking. Oily plastic waste is difficult to clean, so it is frequently placed in landfills or used as solid fuel. The catalyst fast pyrolysis process has gained attention as a method of addressing the challenges posed by plastic waste contaminated with cooking oil. A variety of products, including pyrolysis oil, char and syn gas, can be obtained from this process[3,4].

Numerous researchers have explored the conversion of food-contaminated plastic waste into pyrolysis oil. For example, Pattaraprakorn and Bhasaputra[2] utilized a kaolin catalyst in the pyrolysis process of contaminated food plastic bags, resulting in pyrolysis oil with distinct layers; it was predominantly composed of 69% diesel and 24% gasoline. Wang et al.[5] investigated the catalytic co-pyrolysis of waste vegetable oil and high-density polyethylene, which improved hydrocarbon fuel production. Mahari et al.[6] explored the microwave co-pyrolysis of waste polyolefins and waste cooking oil using activated carbon, obtaining pyrolysis oil with desirable properties for use as fuel in combustion engines.

Nevertheless, wax formation produced from vegetable oil in plastic waste has not gained much attention, despite the fact that it has greater value than pyrolysis oil. The main objective of this study was to investigate the effects of the catalyst and operation conditions of the co-pyrolysis process on the yield and properties of pyrolysis oil and wax.

Materials and methods

Feedstock preparation and reactor start-up

The pyrolysis feed stock was a PP plastic bag obtained from ABC Thai's brand. The PP plastic bag was cut into 2×2 cm^2 pieces and kept under dry conditions before being used in further experiments. The weight of the PP plastic was standardized at five grams throughout the experiments. All experiments were conducted triplicate.

Palm oil was purchased from Sime Darby Oils Morakot Corporation. Activated carbon made from rice husk (RHAC) was obtained from Chemipan Corporation Company, Thailand. Kaolin (46.66% SiO_2, 39.63% Al_2O_3) was purchased from Sigma-Aldrich Corporation.

Catalyst caracterization of kaolin and activated carbon from rice husk

Kaolin and RHAC were characterized to determine their surface and catalytic properties using XRD, XRF, FTIR, BET surface area analysis, and temperature-programmed desorption (TPD) to assess acidic and basic active sites. TPD measurements were conducted using CO_2 and NH_3 as probe gases to evaluate basic and acidic sites, respectively. Prior to analysis, the catalysts were pre-treated with N_2 gas at a flow rate of 60 mL/min for 60 min at 250 °C to remove surface contaminants. For basic site analysis, the pre-treated catalyst was exposed to CO_2 gas (99.9%) at a flow rate of 30 mL/min for 1 h, while NH_3 gas (10% in helium gas) was used for acidic site analysis under the same conditions. After adsorption, excess gas was purged with N_2 at 20 mL/min for 1 h. The desorption step was then carried out by heating the sample from 50 °C to 900 °C under a controlled flow of helium gas.

Total organic content (TOC) analysis was performed on kaolin, RHAC, mixed catalyst and spent catalyst by TOC analyzer (Shimadzu, model TOC-V-CPH) equipped with solid sample analyzer module (model SSM-5000 A).

Experimental setup and operating conditions of the pyrolysis process

A cylindrical quartz reactor was employed in this study, with a diameter of 2.5 cm and a length of 36 cm. The reactor was placed inside an electric furnace measuring 18.3 cm in diameter and 39 cm in height. A glass condenser, 45 cm in length, with an outer diameter of 2.3 cm and an inner diameter of 1.0 cm, was used for vapor condensation. Tap water at 30 °C was circulated through the condenser at a flow rate of 1 L/min to maintain effective cooling.

Five grams of cut palm oil-contaminated PP plastic was put into the quartz reactor before a certain amount of palm oil, activated carbon and kaolin was added. The quartz reactor was put into the electric furnace, which was connected to the air condenser unit at 30 °C. The temperature started at 30 °C with a heating rate of 10–20 °C/min, and the setup temperature range was 400–500 °C. The condensed pyrolysis oil was collected in a 250 ml flask and weighed at 5 min intervals. The process was stopped once there was no condensed pyrolysis oil.

The weighing technique and mass balance were applied to measure the product fractions, including solid, wax, pyrolysis oil and gas. The total mass of the process included the weights of the PP plastic, palm oil and RHAC. The pyrolysis oil fraction was collected and stored in a glass vial at 4 °C for further property analysis.

Characterization of the properties of pyrolysis oil and wax

The properties of the pyrolysis oil product were characterized; namely, its heating value was measured using a bomb calorimeter (ASTM D240), its kinematic viscosity was measured at 40 °C (ASTM D445), and its acid value was measured (ASTM D664). The total energy was normalized by the weight of PP plastic and palm oil, which were the main contributors to pyrolysis oil, as shown in Eq. (1)

$$Total\ Energy\ \left(\frac{KJ}{g\ feedstock}\right) = \frac{(Pyrolysis\ Oil\ (g) \times\ Heating\ Value\ (KJ/g))}{PP\ plastic\ (g) + Palm\ Oil\ (g)} \tag{1}$$

Pyrolysis wax was characterized by its lubricant properties, which included the kinematic viscosity at 40 °C and 100 °C (ASTM D445), viscosity index (ASTM D2270), pour point (ASTN D97) and peroxide value (AOAC (2023) 965.33) for oxidative stability. FTIR analysis was applied to investigate the functional groups involved in wax formation. The TGA profile of wax formation was investigated.

Both pyrolysis oil and wax were analyzed by GC-MS to identify the chemical profile. The components in the pyrolysis oil and wax were identified by comparing their mass spectrogram with those from the mass spectral data library of National Institute of Standards and Technology. The matching degree must be greater than 85% as the selecting criterial. A semi-quantitative method was introduced to determine the relative content of each component in pyrolysis oil and wax based on chromatographic area percentage. It should be noted that percentage cut-off was set at 0.1% of total area.

Statistical analysis

STATISTICA 10 (StatSoft, Tulsa, OK, USA) was used to perform the ANOVA test and multiple mean comparisons at 95% confidence intervals (p values < 0.05).

Results and discussion

Catalyst characterization

The surface properties and chemical compositions of kaolin and RHAC are summarized in Table 1. RHAC exhibited a dramatically higher BET surface area (1,291.88 m^2/g) compared to kaolin (10.85 m^2/g), reflecting its highly porous structure. Similarly, RHAC showed a significantly greater pore volume (0.959 cm^3/g) than kaolin (0.018 cm^3/g). In terms of pore size, RHAC and kaolin exhibited average pore diameters of 2.97 nm and 6.67 nm, respectively, indicating that both materials are predominantly microporous. The acidity and basicity, measured by TPD, further differentiate the two materials. RHAC possessed markedly higher concentrations of both acid site (0.564 mmol/g) and basic sites (6.853 mmol/g), suggesting a more chemically active surface than kaolin with acid site 0.186 mmol/g of and basic site of 1.821 mmol/g.

Regarding chemical composition, RHAC contained a high amount of SiO_2 (38.45%) along with a measurable level of total organic carbon (TOC), whereas kaolin exhibited high levels of both SiO_2 (57.61%) and Al_2O_3 (33.57%), consistent with its natural aluminosilicate structure.

The FTIR spectrum of kaolin (Fig. 1A) exhibits characteristic bands associated with its aluminosilicate structure. The O–H stretching vibrations of structural hydroxyl groups in kaolinite appear at 3673 and 3622 cm^{-1}, confirming the presence of inner and surface hydroxyls. A distinct band at 942 cm^{-1} corresponds to Al–OH bending. The silicate framework is evidenced by the Si–O stretching vibration at 1119 cm^{-1}, along with Si–O symmetric vibrations at 753 and 694 cm^{-1}. The band at 514 cm^{-1} is attributed to Al–O–Si bending, further confirming the layered aluminosilicate structure of kaolin.

In the case of RHAC (Fig. 1B), the FTIR spectrum reveals a variety of functional groups, primarily oxygen-containing groups, and silica-related features. The peaks at 1695 and 1591 cm^{-1} are assigned to C = O stretching and aromatic C = C stretching, respectively, indicating the presence of carbonyl and conjugated structures on the RHAC surface. Strong bands at 1163, 1074, and 1005 cm^{-1} correspond to Si–O–Si and Si–O stretching vibrations, while the band at 875 cm^{-1} is associated with Si–O bending. These signals confirm the presence of residual silica in the RHAC, which is a well-known constituent of rice husk-derived carbon.

The XRD patterns of kaolin, RHAC, and their respective samples after pyrolysis at 500 °C are shown in Fig. 2. The kaolin sample (Fig. 2A) exhibited sharp diffraction peaks corresponding to crystalline phases including SiO_2 (quartz), $Al_2Si_4O_{10}(OH)_2$ (pyrophyllite), and $KAl_2(Si,Al)_4O_{10}(OH)_2$ (muscovite), indicating its well-ordered aluminosilicate structure. In contrast, the XRD pattern of RHAC (Fig. 2C) displayed a broad hump centered around $2\theta = 22–25°$, characteristic of amorphous carbon, with no distinct crystalline peaks observed.

Applying pyrolysis at 500 °C to simulate the conditions used in this study, kaolin (Fig. 2B) retained its original crystalline structure with no significant changes in peak intensity or position. This result suggests that kaolin is thermally stable under pyrolysis conditions and may be reused as a catalyst without structural degradation. On the other hand, RHAC after pyrolysis (Fig. 2D) exhibited a slightly sharper and more pronounced peak near $2\theta = 23°$, indicating the beginning of structural reordering or partial graphitization. This subtle increase

Surface properties	Kaolin	RHAC
BET surface area (m^2/g)	10.85	1,291.88
Pore volume (cm^3/g)	0.018	0.959
Pore size (nm)	6.67	2.97
Acid site from NH_3 desorption (TPD) (mmol/g)	0.186	0.564
Basic site from CO_2 desorption (TPD) (mmol/g)	1.821	6.853
Chemical composition (%)	Kaolin	RHAC
Total organic carbon	Not Detected	57.83
MgO	0.13	0.42
$Al2O_3$	33.57	0.11
SiO_2	57.61	38.45
K_2O	6.81	2.19
TiO_2	0.60	Not detected
Fe_2O_3	0.73	0.06

Table 1. Characterization of Kaolin and RHAC catalyst.

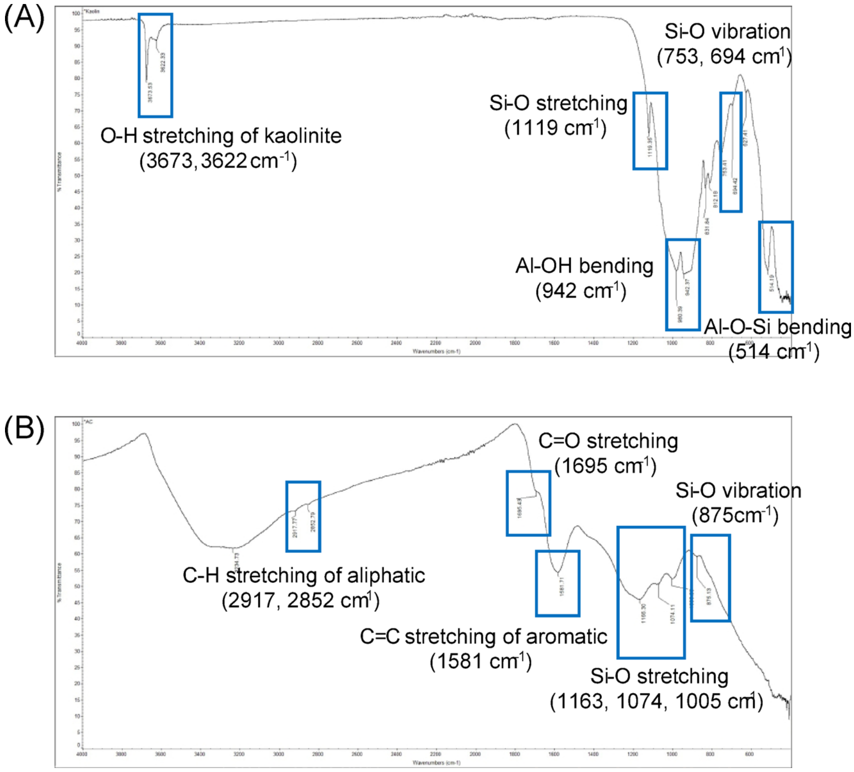

Fig. 1. FTIR spectrum of kaolin (**A**) and RHAC (**B**).

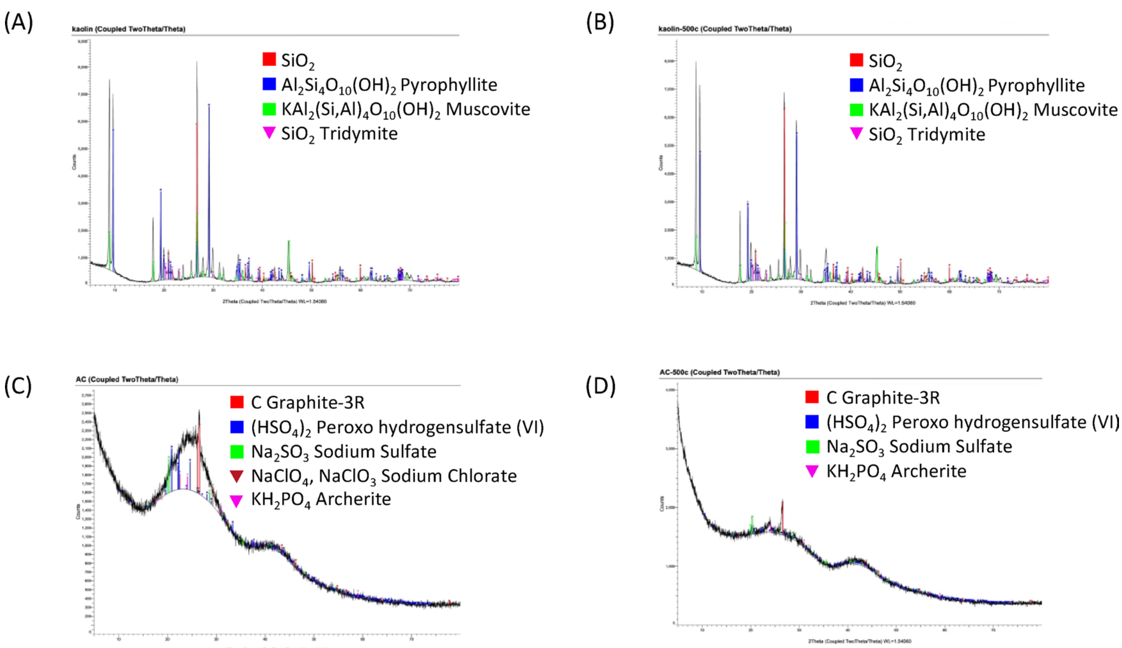

Fig. 2. XRD result of kaolin (**A**), pyrolysis kaolin at 500 °C (**B**), RHAC (**C**) and pyrolysis RHAC at 500 °C (**D**).

in crystallinity may enhance thermal conductivity and electron mobility, potentially improving its catalytic performance.

Notably, no clear SiO_2-related peaks were observed in either RHAC or pyrolyzed RHAC samples. This absence is likely due to the predominantly amorphous nature of silica in rice husk, which does not produce sharp diffraction peaks and is often embedded within the carbon matrix.

In summary, the comprehensive characterization of kaolin and RHAC using FTIR, XRD, XRF, BET, and TPD techniques confirmed their suitability as catalysts for the pyrolysis process. Kaolin offers thermal stability, whereas RHAC provides high surface reactivity and bifunctional acid–base characteristics.

Influence of palm oil on the distributions of product fractions and the properties of pyrolysis oil

The effects of four different ratios of palm oil to PP on the distribution of product fractions and the properties of pyrolysis oil were investigated. The pyrolysis conditions were maintained at 500 °C with a heating rate of 20 °C/min, and 10% kaolin was used as a catalyst. The appearance of the pyrolysis oil produced from each palm oil to PP plastic ratio varied significantly, as illustrated in Fig. 3A. In the absence of palm oil, the pyrolysis oil was clear with a light-yellow color. However, as the ratio of palm oil to PP increased, the oil exhibited greater turbidity and a dark brown color.

With respect to product distribution, increasing the ratio of palm oil to PP markedly increased the liquid fraction, as shown in Fig. 3B. Wang et al.[5] reported similar findings, where the use of waste vegetable oil as a co-pyrolysis feedstock for high-density polyethylene (HDPE) plastic resulted in an increased liquid fraction of the pyrolysis products. Plastics such as HDPE and PP can act as hydrogen donors during the pyrolysis process, which helps suppress the formation of solid residues[7].

In terms of pyrolysis oil properties, the presence of palm oil significantly reduced the heating value compared with the use of PP plastic as the sole feedstock (Fig. 3C). Nevertheless, owing to the increased liquid fraction yield, the total energy per unit of feedstock across different palm oil to PP plastic ratios did not differ significantly. However, increasing the palm oil to PP plastic ratio markedly increased the kinematic viscosity and acid value of the pyrolysis oil, as shown in Fig. 3D.

Palm oil comprises triglycerides, free fatty acids, and various degradation products. During the pyrolysis process, carboxylic acid molecules formed from the co-pyrolysis of vegetable oil and plastic can combine into larger-molecular-weight ketones through ketonization reactions, which may subsequently undergo aldol condensation to form even larger ketones[8]. Therefore, the presence of carboxylic acids derived from the thermal degradation of palm oil during pyrolysis contributes to the increased viscosity and acid value of the resulting pyrolysis oil. Nevertheless, wax formation was not observed in this experiment. This might be due to the secondary degradation at high temperatures and catalyst loadings[9].

On the basis of a preliminary study of residual cooking oil in PP plastic bag waste from food delivery, we found that the ratio of used cooking oil to PP plastic was approximately 1.5 after rinsing with water. Therefore, this palm oil to PP plastic ratio of 1.5 was utilized in subsequent studies.

Influence of the kaolin catalyst on the distributions of product fractions and the properties of pyrolysis oil

The effect of kaolin loading (%wt of PP plastic) on the distribution of pyrolysis products and pyrolysis oil properties was evaluated; the use of only PP plastic was compared with the presence of palm oil at a palm oil to PP plastic ratio of 1.5. The pyrolysis conditions were maintained at 500 °C with a heating rate of 20 °C/min. The results revealed that increasing the kaolin loading improved the pyrolysis oil fraction by reducing the solid fraction (Fig. 4A). The acid active site of kaolin reduced the activation energy, contributing to the dehydration/

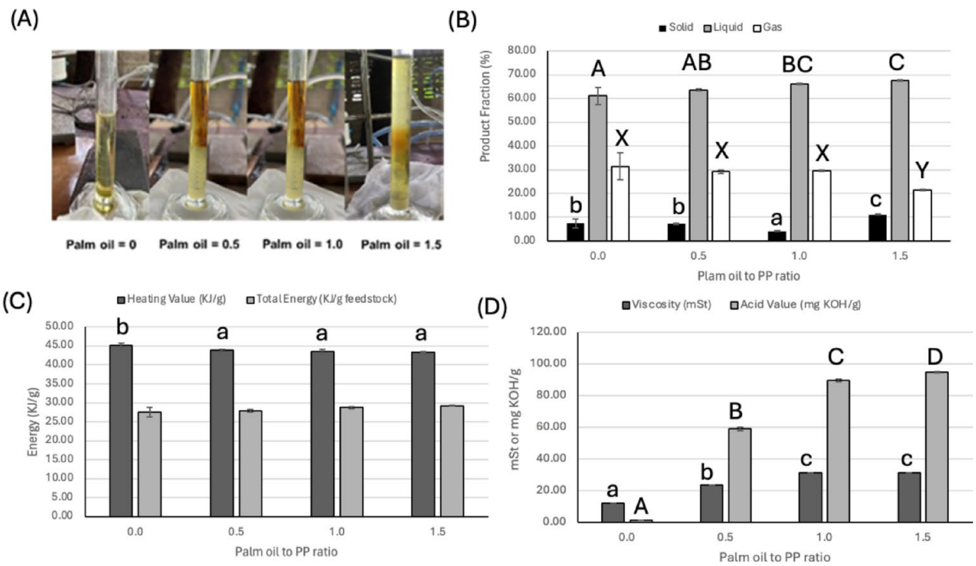

Fig. 3. Effect of palm oil loading on (**A**) pyrolysis oil appearance with (**B**) its production fraction and pyrolysis oil properties, including the (**C**) heating value and total energy and (**D**) kinematic viscosity and acid value at 500 °C with a heating rate of 20 °C/min, using polypropylene (PP) plastic supplemented with 10% kaolin as feedstock. Letters in each figure indicate significant difference in each treatment (p-value < 0.05).

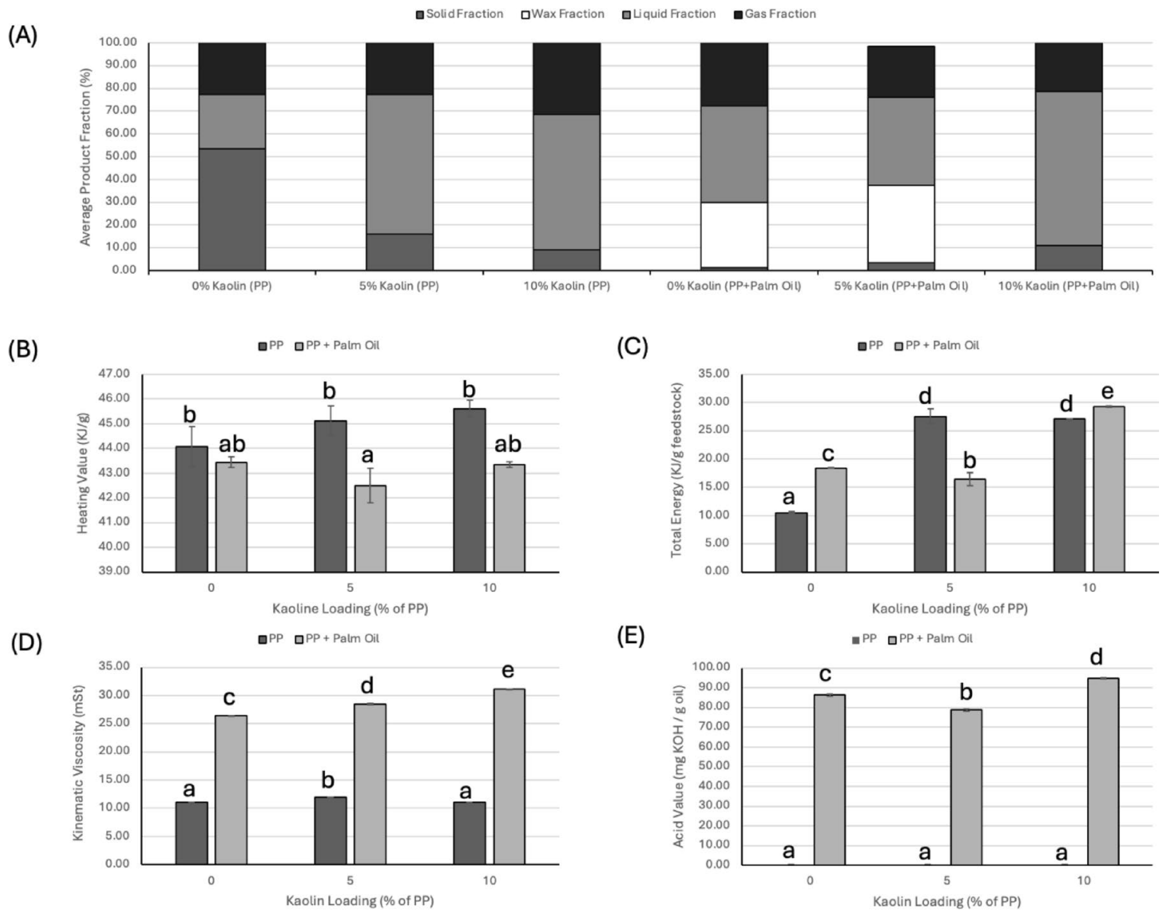

Fig. 4. Effects of kaolin loading and the presence of palm oil on (**A**) the average production fraction of pyrolysis oil and the pyrolysis oil properties, including the (**B**) heating value, (**C**) total energy per feedstock, (**D**) kinematic viscosity and (**E**) acid value of the pyrolysis process at 500 °C with a heating rate of 20 °C/min, using PP plastic and a palm oil to PP plastic ratio of 1.5 as feedstocks. Letters in each figure indicate significant difference in each treatment (p-value < 0.05).

hydrogenation of unsaturated aldol condensation products to form fuel-grade alkanes[10]. Li et al.[11] introduced alkaline-modified HZSM-5 as an effective catalyst for the co-pyrolysis of cotton stalk and polypropylene (PP) plastic. They emphasized that the hydrogen-to-carbon effective ratio (H/Ceff) is a critical factor in determining feedstock performance. The high H/Ceff value of PP, when combined with cotton stalk, enhanced pyrolysis oil yield and reduced coke formation.

Surprisingly, wax formation was observed when the kaolin loading was less than 10% (Fig. 4A). Wax-like products can be mixtures of long-chain hydrocarbons, such as paraffins, olefins, and aromatic or polymerized compounds. In the presence of palm oil, the ketonization of fatty acid-degraded palm oil was observed[9].

However, wax formation can be thermally decomposed under extreme conditions. Jan, Shah, and Gulab[12] reported that the use of a $MgCo_3$ catalyst with HDPE plastic in the pyrolysis process can reduce wax formation. Wang et al.[5] reported that a high concentration of olefin as a hydrogen donor, increasing catalyst loading and optimizing the operation temperature and heating rate could prevent wax formation due to more complete thermal degradation.

With respect to the properties of pyrolysis oil, increasing the kaolin loading did not significantly impact the heating value, both with and without the presence of palm oil (Fig. 4B). However, owing to wax formation at kaolin loadings below 10% in the presence of palm oil, the total energy per unit of feedstock at 10% kaolin loading was greater (Fig. 4C).

Kaolin loading slightly influenced the kinematic viscosity when only PP plastic was used, but it significantly reduced the kinematic viscosity of the pyrolysis oil produced from the cofeeding of PP plastic and palm oil (Fig. 4D). The active sites on kaolin can facilitate reactions between hydrogen donated from PP plastic and carboxylic acids to produce alkanes[13]. Conversely, the acid value data indicated that higher kaolin loading tended to increase the acid value (Fig. 4E). This increase may be attributed to the vapor forms of carboxylic acids or volatile organic acids having insufficient time to react with hydrogen from PP plastic at the kaolin active sites[14]. In addition to increasing the kaolin loading, extending the contact time in the pyrolysis reactor by reducing the heating rate could help lower the acid value, as discussed in section *Influence of heating rate*.

Influence of operation temperature on the distributions of product fractions and the properties of pyrolysis oil

The effects of operating temperature (400, 450, and 500 °C) at a heating rate of 20 °C/min with 10% kaolin loading on the distribution of pyrolysis products and the properties of pyrolysis oil were evaluated, and the results were compared between PP plastic alone and the co-pyrolysis of palm oil at a palm oil-to-PP ratio of 1.5. The results indicated that PP plastic and palm oil exhibited different thermal degradation temperatures. At 400 °C, no pyrolysis oil was observed when PP plastic was used as the sole feedstock. In contrast, with a palm oil to PP plastic ratio of 1.5, up to 50% pyrolysis oil was obtained (Fig. 5A). This might be due to the volatile compounds released from the thermal degradation of palm oil. Wang et al.[5] reported that co-pyrolysis of waste vegetable oil and HDPE at temperatures between 400 and 460 °C favored the production of pyrolysis oil. The discrepancy in the optimal operating conditions in our study may be attributed to the use of different catalysts; they employed ZrO_2-based polycrystalline ceramics, which generally possess more acidic active sites than the kaolin used in this work (Table 1).

Wax formation was observed at operating temperatures below 500 °C in the presence of palm oil. Higher pyrolysis temperatures enhanced hydrogen transfer from PP-derived aliphatic compounds to the thermally degraded products of palm oil, resulting in the formation of alkanes and a reduction in olefin production[13].

In terms of the quality of pyrolysis oil, the operating temperature had a minimal effect on the heating value, both with and without the presence of palm oil (Fig. 5B). The total energy per unit of feedstock was influenced primarily by wax formation when PP plastic and palm oil were used as co-feedstocks (Fig. 5C). At 400 °C, the pyrolysis oil produced from PP plastic in the presence of palm oil presented the highest kinematic viscosity; however, this viscosity decreased as the temperature increased to 500 °C (Fig. 5D). This reduction may be attributed to insufficient energy to promote the formation of alkanes from the PP plastic as hydrogen donors and carboxylic acids derived from the thermal degradation of palm oil[5]. These findings were supported by the observed decrease in the acid value of the pyrolysis oil with increasing operating temperature (Fig. 5E).

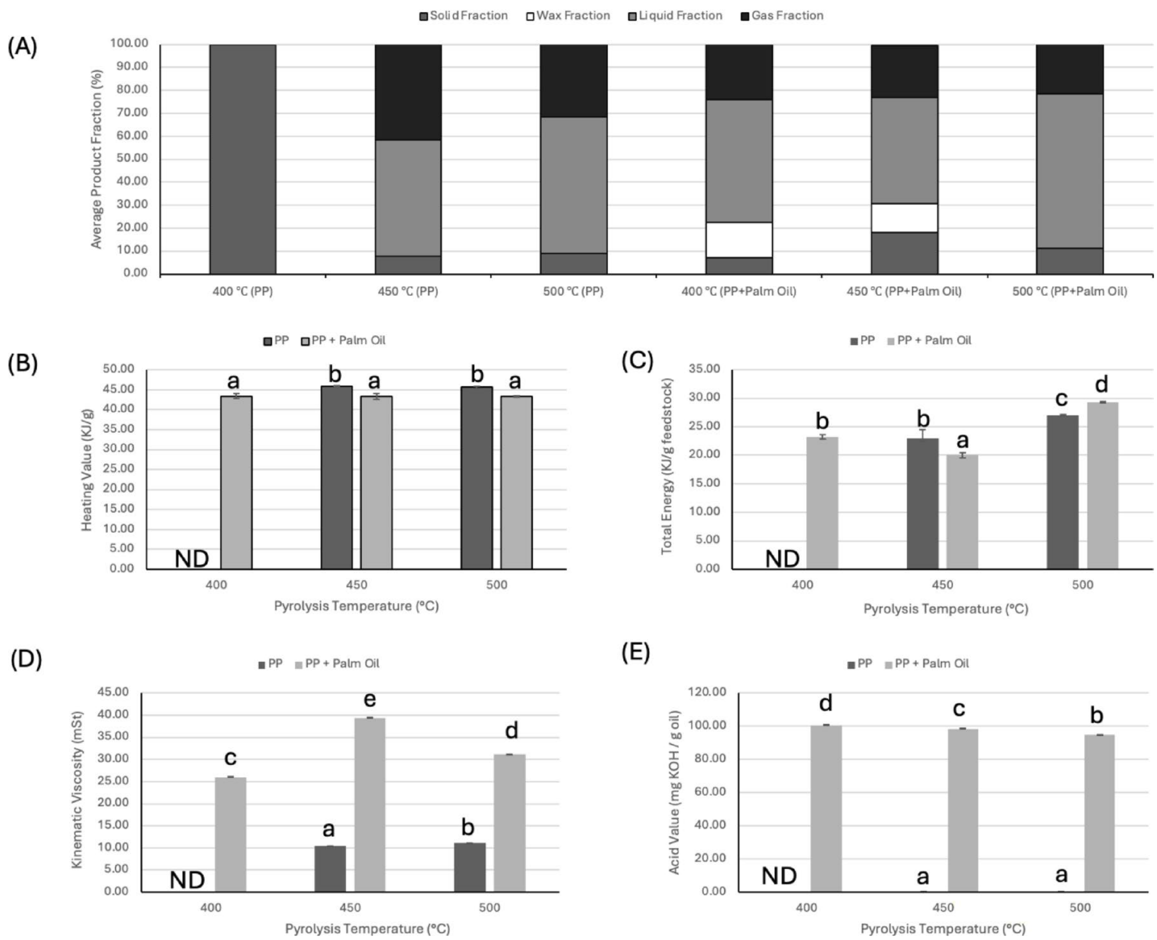

Fig. 5. Effects of pyrolysis temperature and the presence of palm oil on (**A**) the average production fraction and pyrolysis oil properties, including the (**B**) heating value, (**C**) total energy per feedstock, (**D**) kinematic viscosity and (**E**) acid value at a heating rate of 20 °C/min, when PP plastic and a palm oil to PP plastic ratio of 1.5 (both supplemented with 10% loading of kaolin) were used as feedstocks. Letters in each figure indicate significant difference in each treatment ($p\text{-value} < 0.05$).

Influence of heating rate on the distributions of product fractions and the properties of pyrolysis oil

The effect of the heating rate at 500 °C with 10% kaolin on the distribution of pyrolysis products and the properties of pyrolysis oil was examined, and the results were compared between the PP plastic alone and the co-pyrolysis of palm oil at a palm oil to PP plastic ratio of 1.5. Consistent with findings related to kaolin loading and operating temperature, the highest heating rates inhibited wax formation and increased the yield of pyrolysis oil (Fig. 6A). Mahari et al.[6] reported that a higher heating rate results in shorter reaction times and lower energy consumption during the pyrolysis process, thus not significantly affecting the oil yield. However, in our study, insufficient retention time during pyrolysis led to wax formation, which subsequently reduced the fraction of pyrolysis oil, as illustrated in Fig. 6A.

The heating rate did not significantly affect the heating value of pyrolysis oil in the presence of palm oil; however, it had a pronounced effect on the heating value of pyrolysis oil produced solely from PP plastic as the feedstock (Fig. 6B). At a heating rate of 15 °C/min, the lowest heating value was observed. This may be attributed to the thermal cracking of PP-derived aliphatic compounds into smaller molecules, such as methane and hydrogen gas. The increased gas fraction produced at this heating rate supports this observation (Fig. 6A). Future studies should consider investigating the properties and profile of the pyrolysis gas generated.

Although the heating rates of pyrolysis oil at each heating level with the presence of palm oil showed no significant differences (Fig. 6B), the highest heating rate of 20 °C/min yielded the greatest total energy (Fig. 6C). This increase can be attributed to the higher proportion of pyrolysis fractions, likely resulting from the degradation of wax formation.

Increasing the heating rate significantly elevated the kinematic viscosity and acid value of the pyrolysis oil produced from PP plastic in combination with palm oil (Fig. 6D and E). This increase may be attributed to insufficient residence time in the reactor, which limited the opportunity for carboxylic acids derived from the thermal degradation of palm oil to react with the hydrogen released from PP plastic at the active site of the kaolin catalyst in the reactor.

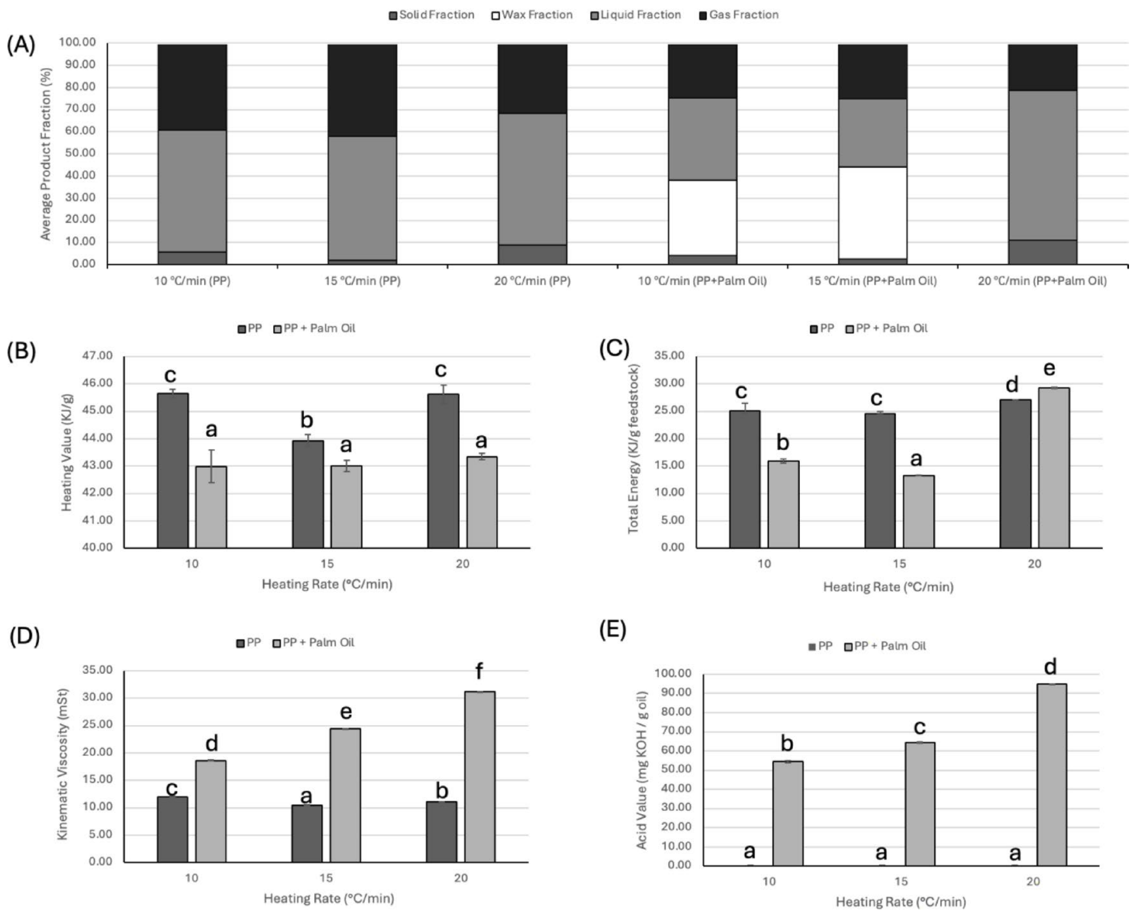

Fig. 6. Effects of the heating rate and presence of palm oil on the (**A**) average production fraction and pyrolysis oil properties, including the (**B**) heating value, (**C**) total energy per feedstock, (**D**) kinematic viscosity and (**E**) acid value at a pyrolysis temperature of 500 °C, when PP plastic and a palm oil to PP plastic ratio of 1.5 (both supplemented with 10% loading of kaolin) are used as feedstocks. Letters in each figure indicate significant difference in each treatment (p-value < 0.05).

Influence of RHAC on the distributions of product fractions and the properties of pyrolysis oil

The results indicated that the use of RHAC promoted wax formation (Fig. 7A). Qiu et al.[15] investigated rice husk biochar as a pyrolysis catalyst and noted that it contains abundant N- and O- functional groups, along with a high density of acidic sites. These surface properties can facilitate oligomerization, leading to increased wax production. Premchand et al.[16] evaluated the hydrocarbon product from the co-pyrolysis of walnut shells and palm oil fatty acid distillate using sugarcane bagasse biochar with N, P, O and B doping. The results illustrated that the addition of biochar, N-doped biochar, O-doped biochar and P-doped biochar promoted the selectivity of longer-chain hydrocarbons by reducing shorter-chain hydrocarbons.

The addition of activated carbon as a cocatalyst had a minor effect on the heating value of pyrolysis oil produced from PP plastic in the presence of palm oil. However, at a 10% loading, activated carbon significantly reduced the heating value of the pyrolysis oil derived from only the PP plastic (Fig. 7B). Surprisingly, applying 20% RHAC promote the wax formation (Fig. 7A), resulting in reducing in total energy (Fig. 7C).

In terms of kinematic viscosity, activated carbon had no noticeable effect (Fig. 7D). When PP plastic was combined with palm oil, the inclusion of RHAC significantly lowered the acid value (Fig. 7E). This reduction is attributed to the abundance of acidic active sites on the surface of RHAC, which can promote decarbonylation and esterification[15], resulting in a lower concentration of carboxylic acids in the pyrolysis oil.

Based on the product distribution and properties of pyrolysis oil, the subsequent experiments were conducted using the pyrolysis oil and wax produced at 500 °C with a heating rate of 20 °C/min, utilizing palm oil to PP ratio of 1.5, both supplemented with 10% kaolin loading.

Reusing catalyst

The selected conditions for producing pyrolysis oil and wax from palm oil-contaminated PP plastic involved using 5 g of PP plastic mixed with 7.5 g of palm oil (a 1.5:1 palm oil to PP ratio) as the feedstock. A catalyst mixture consisting of 0.5 g of kaolin (10% wt of PP) and 1 g of RHAC (20% wt of PP) was used. The pyrolysis was carried out at 500 °C with a heating rate of 20 °C/min.

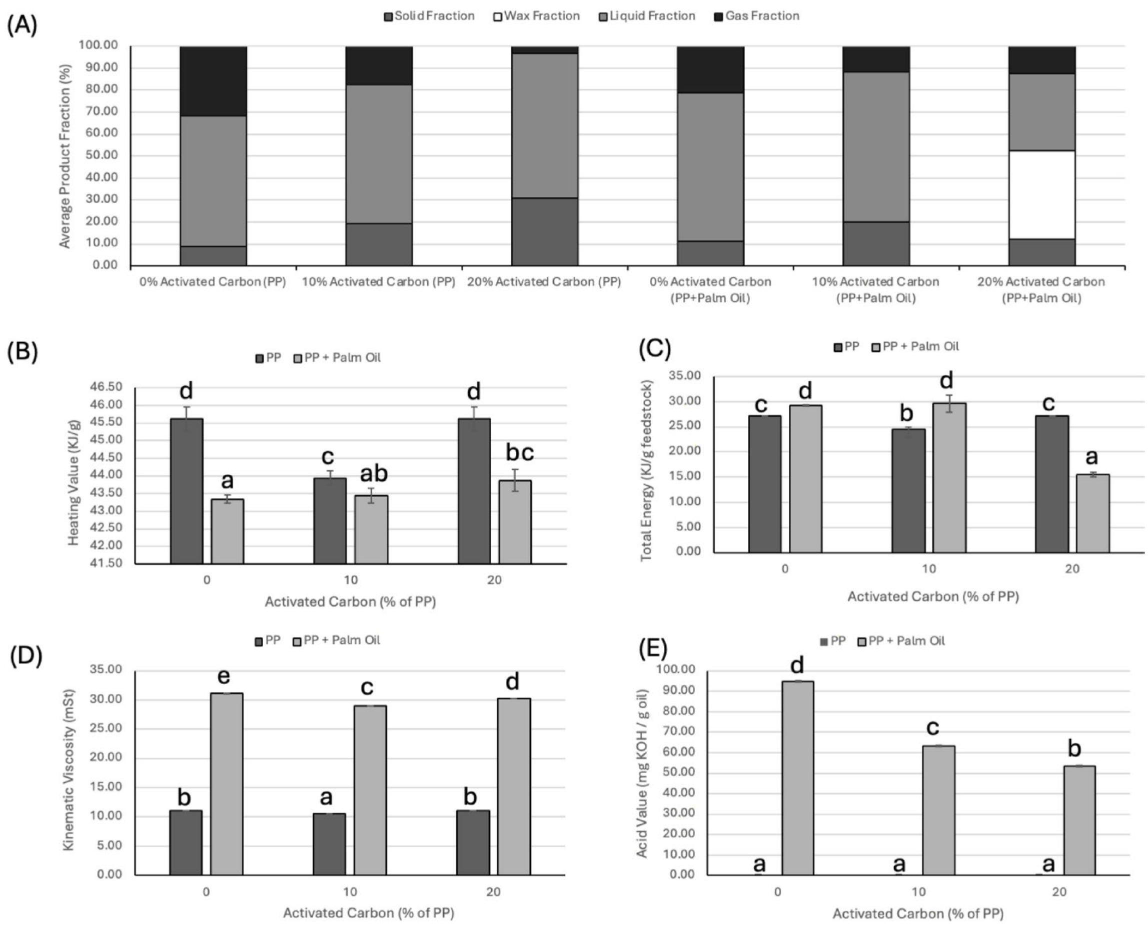

Fig. 7. Effects of RHAC and the presence of palm oil on the (**A**) average production fraction and pyrolysis oil properties, including the (**B**) heating value, (**C**) total energy per feedstock, (**D**) kinematic viscosity and (**E**) acid value at a pyrolysis temperature of 500 °C with a heating rate of 20 °C/min, when PP plastic and a palm oil to PP plastic ratio of 1.5 (both supplemented with 10% loading of kaolin) were used as feedstocks. Letters in each figure indicate significant difference in each treatment (p-value < 0.05).

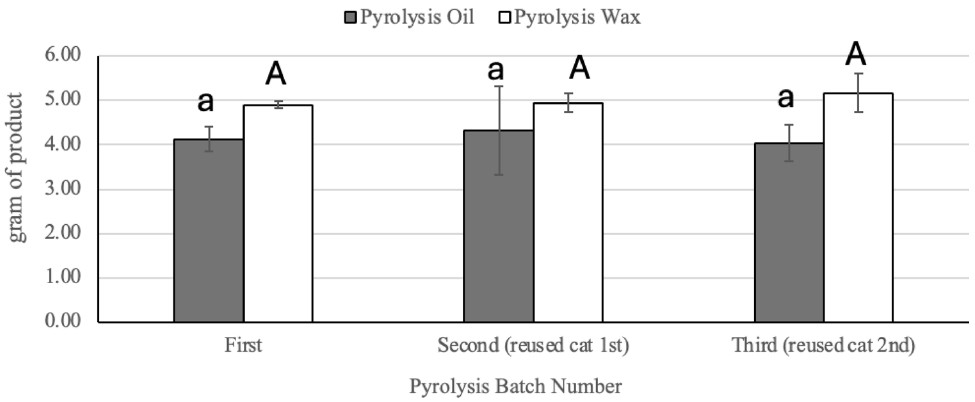

Fig. 8. Weight of produced pyrolysis oil and pyrolysis wax from fresh catalyst, reused spent catalyst 1 and 2 times under pyrolysis temperature of 500 C using 12.5 g of mixed PP-palm oil as feedstock. Letters in each figure indicate significant difference in each treatment (p-value < 0.05).

Single catalyst	**TC (%)**	**IC (%)**	**TOC (%)**
Kaolin	0.25	0.17	0.08
Activated carbon	58.07	0.23	57.83
Mixed catalyst	**TC (%)**	**IC (%)**	**TOC (%)**
New catalyst (1st batch)	26.94	0.20	26.73
Spent catalyst (2nd batch)	60.76	0.25	60.51
Spent catalyst (3rd batch)	50.78	0.28	50.50

Table 2. TOC analysis of catalyst and spent catalyst used in pyrolysis process.

In the first batch, fresh catalyst was used. The spent catalyst from this run was reused in the second batch without any regeneration or pretreatment. Similarly, the catalyst from the second batch was used in the third batch under the same conditions. Pyrolysis oil and wax were collected from each batch, and their weight and chemical compositions were analyzed using GC-MS to assess the reusability and performance of the developed catalyst.

The results showed that there was no significant difference in the yields of pyrolysis oil and wax when the spent catalyst was reused for two consecutive pyrolysis cycles (Fig. 8). This stability in performance is likely due to the initially high catalyst loading. However, TOC analysis (Table 2) indicated increased carbon accumulation on the spent catalyst, suggesting gradual deactivation. Interestingly, the TOC of the catalyst used in the third batch was lower than that of the second batch, and the standard deviation in pyrolysis oil yield was also reduced. This observation implies that the thermal conditions during the third pyrolysis cycle may have contributed to partial in situ regeneration of the catalyst. Therefore, controlled thermal regeneration could be a promising approach to sustain catalyst activity over multiple cycles.

Sun et al.[17] emphasized that thermal regeneration of biomass-derived catalysts requires precise control of oxygen levels to prevent over-oxidation. In their study, thermal oxidation was applied to restore the catalytic activity of biochar produced from corn straw. They reported that introducing oxygen at an optimal concentration of 0.4% for 7 min enhanced graphitization and effectively removed 60–80% of the microporous surface blockages. Under these conditions, the catalyst achieved a maximum activity recovery rate of 79.1%.

Similarly, our XRD results (Fig. 2D) indicate that RHAC subjected to pyrolysis at 500 °C exhibited increased graphitization compared to pristine RHAC (Fig. 2C), suggesting that thermal treatment under controlled conditions can enhance catalytic properties through structural modification. Based on these findings, future studies should investigate partial thermal oxidation under limited oxygen conditions as a potential strategy for regenerating catalyst activity in our system.

Chemical profile of pyrolysis oil and wax by GC-MS

Even the yield of pyrolysis oil and wax was not affected by reusing catalyst, the chemical profile of pyrolysis oil and wax of each batch were slightly different.

Pyrolysis oil

The GC-MS analysis revealed that the primary chemical constituents in the pyrolysis oil obtained from both fresh and spent catalysts were alkanes, followed by alkenes and alcohols, respectively (Table 3). Although the pyrolysis oil exhibited a high acid value of 44 mg KOH/g oil, carboxylic acid compounds were not directly

Chemical compounds	Pyrolysis oil (freash catalyst)		Pyrolysis oil (spent catalyst)	
	% Area	No. of peak	% Area	No. of peak
Alcohols	18.23	17	15.07	11
Alkane	15.92	36	12.12	21
Alkene	28.10	42	26.18	28
Amide	0.75	1	0.80	1
Aromatics	2.21	3	2.2	3
Carboxylic acids	N.D.	N.D.	N.D.	N.D.
Ester	3.2	11	0.67	1
Ketone	1.01	3	0.44	1

Table 3. Chemical compounds found in pyrolysis oil produced from fresh catalyst and spent catalyst.

Chemical compounds	Pyrolyis wax (freash catalyst)		Pyrolysis wax (spent catalyst)	
	% Area	No. of peak	% Area	No. of peak
Alcohols	5.76	15	18.48	13
Alkane	9.97	31	8.98	17
Alkene	19.16	48	12.8	18
Amide	0.62	1	0.74	1
Aromatics	0.45	8	N.D.	N.D.
Carboxylic acids	47.99	17	46.19	11
Ester	1.81	1	1.12	2
Ketone	0.76	4	1.31	2

Table 4. Chemical compounds found in pyrolysis wax produced from fresh catalyst and spent catalyst.

detected in either sample. However, several unidentified oxygenated compounds were observed, which may have contributed to the elevated acid value.

When compared with pyrolysis oil derived from biomass waste, the acid value of our pyrolysis oil was relatively lower. For instance, Sundqvist, Oasmaa, and Koskinen[18] reported an acid value of 105 mg KOH/g oil from forest thinning-derived pyrolysis oil. Xu, Jiang, Zhang, and Dai[19] obtained a value of 67 mg KOH/g oil from used cooking oil, while Li et al.[20] reported 86.9 mg KOH/g oil from rubber seed oil. These studies employed esterification reactions to convert carboxylic acid compounds into esters using alcohols, resulting in a significant reduction in acid values to below 10 mg KOH/g oil.

In contrast, since no carboxylic acids were identified in our pyrolysis oil, alternative strategies such as solvent extraction with methanol or distillation may be required to reduce the acid content.

Pyrolysis wax

The composition of pyrolysis wax varied between samples derived from fresh and spent catalysts. Based on peak area analysis (Table 4), carboxylic acids were the predominant compounds, accounting for nearly 50% of the total area, followed by alkenes, alcohols, and alkanes. Palmitic acid and stearic acid were the major constituents among the carboxylic acids.

The FTIR results (Fig. 9) confirmed that the produced wax exhibited a C = O stretching peak at 1715 cm^{-1}, indicating the presence of a carboxylic acid group, and a C = C stretching peak at 1450 cm^{-1}, suggesting the presence of an aromatic ring. Additionally, the characteristic peak of unsaturated fatty acids from the palm oil feedstock at 3010 cm^{-1} disappeared in the wax product, confirming the breakdown of double bonds through the thermal cracking process.

The TGA results (Fig. 10) indicated that the pyrolyzed wax exhibited a broad mass loss between 250 °C and 330 °C, which can be attributed to the diverse molecular structures formed during the thermal cracking of the palm oil and plastic. This is supported by the FTIR and GC-MS analysis, which revealed the presence of various functional groups in the pyrolyzed wax. On the basis of the TGA and DSC data, the approximate boiling point of pyrolysis wax lies within the range of 250 °C to 330 °C. Owing to its thermal properties, pyrolysis wax may not be suitable for high-temperature applications. However, the viscosity index (VI) of the pyrolysis wax exceeded 400, suggesting that its viscosity is relatively stable and less sensitive to temperature changes compared with that of palm oil (Table 5).

The major long chain saturated fatty acid in our pyrolysis wax is commonly used as additives in lubricant applications. Li et al.[21] demonstrated that the addition of 0.3% stearic acid to mineral oil 2137 significantly improved its lubricating properties, resulting in a 13.3% reduction in the coefficient of friction. The high content of oxygenated compounds in the pyrolysis wax raises concerns regarding oxidative stability. However, the measured peroxide value, a common indicator of oxidative stability, was 0.68 meq O_2/kg, which is significantly

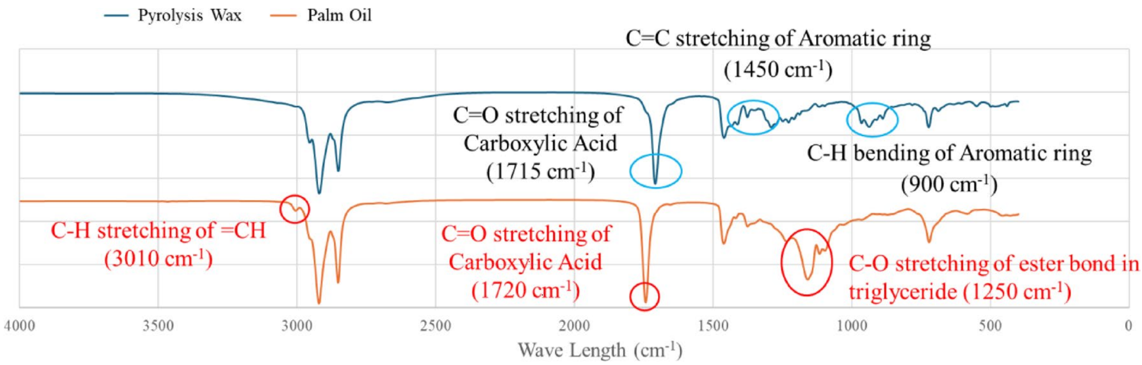

Fig. 9. FTIR of palm oil (red) and pyrolysis wax (blue) produced at a pyrolysis temperature of 500 °C with a heating rate of 20 °C/min using PP plastic and a palm oil to PP plastic ratio of 1.5 (both supplemented with 10% loading of kaolin and 20% loading of activated carbon) as feedstocks.

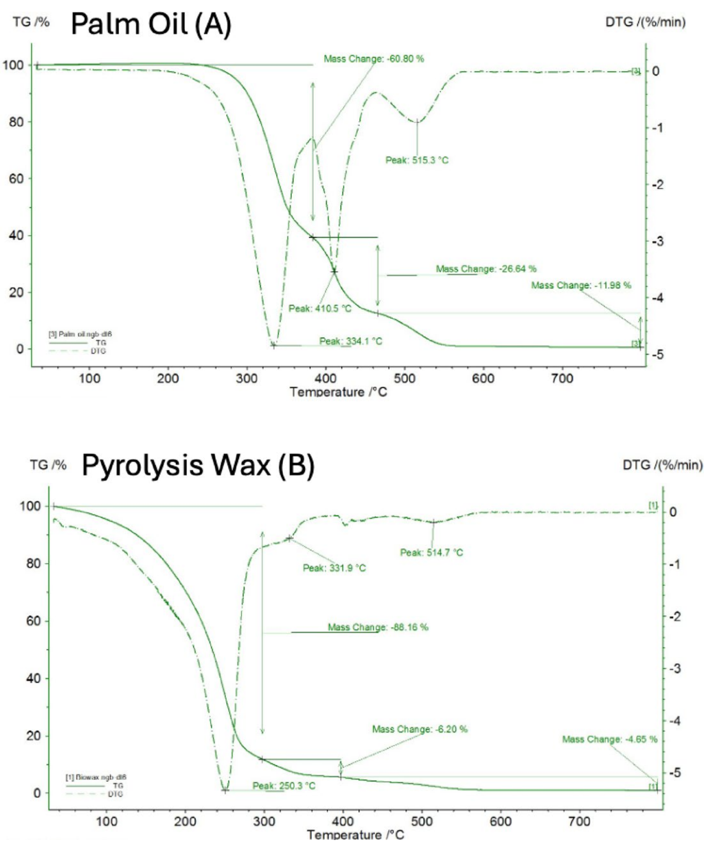

Fig. 10. TGA results under N_2 conditions for palm oil (**A**) and pyrolysis wax (**B**) produced at a pyrolysis temperature of 500 °C with a heating rate of 20 °C/min using PP plastic and a palm oil to PP plastic ratio of 1.5 (both supplemented with 10% loading of kaolin and 20% loading of activated carbon) as feedstocks.

lower than that of palm oil (approximately 2 meq O_2/kg)[22]. This suggests that the pyrolysis wax exhibits better oxidative stability than palm oil. Nonetheless, the high concentration of carboxylic acids contributes to an elevated acid value, indicating the need for further refining processes to improve its quality for practical applications.

Fatty acids present in pyrolysis wax show great potential as feedstock for bio-lubricant production. Various chemical conversion pathways can be applied, including esterification, epoxidation, ketonization, and hydrogenation[23]. In addition to utilizing fatty acids, the use of heterogeneous catalysts is recommended to enhance reaction efficiency and product selectivity in bio-lubricant synthesis[24]. For instance, Li et al.[20] employed a ZrO_2/SBA-15 heterogeneous catalyst to esterify pyrolysis oil, while Rao et al.[25] developed a biochar-based heterogeneous catalyst derived from *Prosopis juliflora* biomass for esterification. Goculdas et al.[26] utilized MgO as a catalyst for the ketonization of lauric acid. Similarly, Hong et al.[27] developed a TiO_2-based catalyst for

Lubricant	Viscosity 40 °C (cSt)	Viscosity 100 °C (cSt)	Viscosity Index	Pour Point (°C)
Mineral oils				
ISO VG32	> 28.8	> 4.1	> 90	-6
ISO VG46	> 41.4	> 4.1	> 90	-6
ISO VG68	> 61.4	> 4.1	> 196	-6
ISO VG100	> 90.0	> 4.1	> 216	-6
Paraffin VG45	95	10	102	-
Paraffin VG460	461	31	97	-
R150	150	-	-	-
SAE20W40	105	13.9	132	-21
AG100	216	19.6	103	-18
75 W-90	120	15.9	140	-48
75 W-140	175	24.7	174	-54
80 W-140	310	31.2	139	-36
This work				
Palm oil	40.16	7.79	161.18	7
Wax	10.5	9.34	404.06	26

Table 5. Physiochemical properties of the mineral lubricant and pyrolysis wax used in this work.

ketonization, followed by hydrotreating using a Pt/Al_2O_3 catalyst to produce long-chain paraffins from palm kernel fatty acid distillate.

Propose chemical reaction occurred in pyrolysis reactor using mixed catalyst

During pyrolysis under limited oxygen conditions, thermal cracking occurs as the primary decomposition pathway. For plastics, the thermal pyrolysis mechanism typically involves β-chain scission initiated by free radicals. In this process, a free radical breaks two carbon atoms away from the charged carbon, forming an olefin (e.g., ethylene) and a primary radical with two fewer carbon atoms[28]. In the case of PP plastic, the polymer chains undergo random scission, breaking down into smaller hydrocarbon molecules that transition from solid to liquid and gas products. When palm oil is co-fed with PP plastic, an additional reaction occurs, which is hydrolysis of triglycerides into free fatty acids and glycerol. This was evidenced by the high acid value in the liquid fraction of the product (Fig. 4E), confirming the formation of carboxylic acid compounds.

Beyond β-scission, Harmon et al.[29] proposed a detailed mechanistic model for PE and PP pyrolysis, which includes multiple reactions: chain fission, allyl chain fission, radical recombination, disproportionation, end-chain and mid-chain β-scission, radical addition, hydrogen abstraction, and β-scission to low molecular weight products. Among these, radical addition, the reverse of β-scission, contributes to oligomerization of short-chain alkenes, which can further cyclize into aromatic compounds such as benzene[30].

The addition of kaolin catalyst during pyrolysis can shift product distribution by facilitating carbocation-mediated chain scission. According to Cai et al.[31], kaolin and montmorillonite clays yielded the highest pyrolysis oil outputs due to their acidic active sites (6.1 and 10.6 mmol/g, respectively). In contrast, the kaolin used in this study had a lower acid site density (0.186 mmol/g; Table 1). However, kaolin still contributes catalytically by lowering the activation energy for chain cleavage, thereby enhancing free radical generation. Luo and teams[32] further demonstrated that acid-treating kaolin with hydrochloric acid increased its active sites, improving pyrolysis oil and gas yields. The acid treated kaolin surface, rich in Al–O and Si–O bonds, also produced oil fractions resembling gasoline and diesel. In the case of palm oil pyrolysis, the basic active sites of kaolin play a key role in promoting hydrolysis of triglycerides into free fatty acids. These free fatty acids can subsequently undergo thermal decarboxylation, yielding long-chain alkanes and alkenes. Further chain scission reactions then convert these into smaller hydrocarbon molecules.

Compared to kaolin, RHAC exhibited a significantly higher density of both acid and basic sites (Table 1). This dual functionality makes RHAC effective in promoting both cracking and hydrolysis reactions. Moreover, its high surface area and graphitized surface, as confirmed by BET and XRD analyses, enhance thermal conductivity and heat transfer. This is especially beneficial in a batch-type reactor without mixing, such as the one used in this study[30]. also reported that activated carbon derived from coconut husk improved pyrolysis efficiency by facilitating rapid and uniform heat distribution, leading to more desirable product yields.

Based on GC-MS analysis, no carboxylic acid compounds were detected in the pyrolysis oil. Instead, alkanes, alkenes, and alcohols were the primary constituents (Table 3). In contrast, the pyrolysis wax contained approximately 50% of its total peak area attributed to palmitic acid and stearic acid, indicating that ketonization reactions were minimal under the current operating conditions (Table 4).

Considering the chemical composition of the pyrolysis products and the catalytic properties of kaolin and RHAC, we hypothesize the following reaction pathway: the basic sites on kaolin and RHAC catalyze the hydrolysis of palm oil triglycerides into free fatty acids and glycerol. The high surface area and graphitized texture of RHAC facilitate efficient heat transfer, minimizing secondary cracking of the long-chain fatty acids. Consequently, the

Inventory	EPPO[33] (USD/kg waste)	Our study (USD/kg waste)	Applying continueous reactor and reusing catalyst in our study (USD/kg waste)
Infrastructure investment	0.245	0.245	2.520
PE/PP plastic	0.196	0.035	0.035
Energy cost	0.196	0.196	0.137
Chemical cost	0.002	0.002	0.002
Catalyst cost	0.015	0.303	0.101
Water fee	0.0002	0.0002	0.0002
Logistic cost	0.049	0.049	0.049
Maintianant cost	0.015	0.015	0.015
Labor cost	0.098	0.098	0.098
Management fee	0.054	0.054	0.054
Total (exclude infrastructure)	0.625	0.753	0.492
Net total	0.870	0.998	3.012
Product			
Pyrolysis oil	0.318	0.194	0.194
Pyrolysis wax	0.000	0.384	0.384
Total	0.318	0.578	0.578

Table 6. Estimating techno-economic feasibility of pyrolysis oil and wax production from plastic waste.

pyrolysis wax becomes enriched in saturated fatty acids (e.g., palmitic and stearic acids), which are polar and readily phase-separate from the hydrophobic pyrolysis oil fraction dominated by alkanes and alkenes.

Techno-economic feasibility of producing pyrolysis oil and wax from cooking oil contaminated PP plastic waste

To evaluate the viability of the developed catalytic pyrolysis process, a techno-economic feasibility assessment was conducted. This evaluation was based on a prior study by the Energy Policy and Planning Office (EPPO) of Thailand[33], which analyzed the economic potential of catalytic pyrolysis for plastic waste. The functional unit of the assessment was defined as one kilogram of plastic waste. The plastic waste was fed into this process approximately 2,166 tons/year. The investment of infrastructure was calculated based on 15 years project period with 6.5% interest rate. Cost parameters from the EPPO study were adapted for this analysis, with adjustments made to account for differences in feedstock and catalyst (Table 6).

While the EPPO study used clean PE/PP plastic waste priced at 0.118 USD/kg, this study utilized cooking oil-contaminated PP plastic waste at a lower cost of 0.035 USD/kg, based on the local RDF (refuse-derived fuel) price. For the catalyst costs, our analysis applied market prices of 3 USD/kg for kaolin and 1 USD/kg for RHAC, reflecting the cost of the developed catalyst formulation. Regarding product yield, the EPPO study reported the production of 600 mL of pyrolysis oil per kilogram of PE/PP plastic. In comparison, our process yielded 0.367 L of pyrolysis oil and 0.435 L of pyrolysis wax per kilogram of contaminated PP plastic. Market fuel prices were used to estimate product value, which were 0.529 USD/L for pyrolysis oil (equivalent to fuel oil) and 0.882 USD/L for pyrolysis wax (based on the price of low-grade mineral oil).

The results presented in Table 6 indicate that neither the EPPO study nor our current study achieved techno-economic feasibility under the evaluated conditions. Although our process successfully produced high-value pyrolysis wax from low-cost, cooking oil-contaminated plastic waste, the revenue generated could not offset the operational costs, excluding infrastructure investment. This outcome was primarily influenced by the market prices of fuel oil and mineral oil used in the economic assumptions.

In Thailand, there have been policy-level efforts to support plastic-to-fuel pyrolysis. For instance, in the EPPO study conducted in 2015[33], the Thai government subsidized the price of pyrolysis oil produced from plastic waste based on the Dubai crude oil benchmark. However, this subsidy was only temporary and did not support long-term economic sustainability. To make the process viable, improvements in energy efficiency and reductions in catalyst cost are necessary.

Typically, batch-type reactors are used in pyrolysis operations due to their low capital investment and suitability for small-scale applications. However, transitioning to a continuous reactor system can reduce energy consumption by up to 70%[34]. This benefit comes at the cost of significantly higher capital investment with approximately 10.28 times greater than that of batch reactors[35], resulting in a longer payback period as a trade-off.

In our study, the developed catalyst demonstrated reusability for up to three cycles, effectively lowering the per-use catalyst cost. When factoring in both the use of a continuous reactor and catalyst reusability, the economic analysis showed that the value of pyrolysis products reached 0.578 USD/kg of plastic waste, while the operational cost was reduced to 0.492 USD/kg. This resulted in a net profit of approximately 0.086 USD for every kilogram of cooking oil-contaminated plastic waste processed.

Nevertheless, the profit margin achieved in this study remains insufficient. Based on an estimated infrastructure investment cost of 2.520 USD per kilogram of plastic waste processed (Table 6), the projected profit translates to a return of only 3.41%, which falls short of the typical interest rate benchmark of 6.5% used in

EPPO study. To enhance the economic feasibility of this pyrolysis project, access to concessional financing, such as soft loans through green finance initiatives, should be considered.

Conclusion

This study demonstrated the feasibility of simultaneously producing pyrolysis oil and wax from cooking oil-contaminated PP plastic using a mixed catalyst system of kaolin and RHAC, which effectively promoted wax formation. However, further upgrading is required for both products. The high oxygenated content in pyrolysis oil suggests the need for refinery process such as distillation. The pyrolysis wax, which was rich in carboxylic acids, can serve as a promising feedstock for bio-lubricant production via esterification, epoxidation, ketonization, and hydrogenation.

Catalyst reusability tests confirmed stable performance over three consecutive cycles, with no significant reduction in product yields and only minor variations in chemical composition. Techno-economic analysis revealed that co-producing pyrolysis oil and wax provided greater economic potential compared to producing oil alone. To enhance the viability of this approach, access to concessional financing mechanisms, such as soft loans from green finance, should be pursued.

Data availability

The data will be made available upon request. Please contact correspondent author (Nattapong Tuntiwiwattanapun; Nattapong.t@chula.ac.th).

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Author contributions
T.S.: conceptualization, methodology, formal analysis, investigation, writing - original draft. V.K.: writing - review & editing, supervision. P.P.: formal analysis, investigation. T.P.: formal analysis, investigation. N. T.: conceptualization, methodology, formal analysis, writing - review & editing, supervision, project administration.

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Declarations

Consent to participate
All the authors provided consent to participate in this study.

Competing interests
The authors declare no competing interests.

Additional information
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